OPERATIONS RESEARCH.

⇒ Operations Research (OR) is the application of scientific and mathematical methods to study and analyze problems involving complex systems for decision making, optimization, and efficiency.

1. Critically analyze how the collaboration between managers and OR specialists influences the effectiveness of OR models. ~~Po.~~ Provide real or hypothetical examples.

⇒ The collaboration between managers and operations research (OR) specialists is crucial in determining the effectiveness and real-world applicability of OR models.

- While OR specialists brings technical expertse in modeling, optimization and analytics, managers contribute domain knowledge, strategic insights and access to organizational data and constraints.

→ when collaboration is effective, OR models are more accurate, relevant and implementable; when it fails, models risk being underutilized or outright reject.

1. Complementary ~~Ana~~ Expertise

- OR specialists excell in creating mathematical model and simulations.
- Managers understands the business context, goals and constraints.

If both parties communicate effectively, models are tailored to actual problems rather than theoretical ideals.

Ex:-

In a manufacturing firm, OR specialists might design a linear programming model to optimize production schedules. If managers are involved, they can ensure the model includes real-world limitations such as ~~mana~~ machine maintenance windows or labor union constraints. This makes the solutions implementable rather than just optimal on paper.

2. Risk of Misalignment

without proper collaboration:

- OR specialists may misdefine the problem.
- Managers may misinterpret results or lack trust in the model.

Hyphothetical case:

A retail chain wants to optimize inventory across outlets. The OR team develops a model minimizing holding and transportation costs. However, they fail to consider seasonal demand patterns, known only to experienced regional managers. The model performs poorly, and the project is scrapped.

3. Iterative feedback Improves Model validity.

frequently interactions allows:

- Refinement of assumptions
- Validation with actual data.
- Rapid response to model weakness

4. Implementation challenges

Even the best model can fail if managers:

- Are not trained to use it.
- Lack buy-in due to poor involvement during development.
- Fear that models will replace human decision-making.

5. Success Relies on Shared Ownership

when managers co-develop the model:

- They're more likely to trust and use it.
- Model recommendations aligns with strategic goals.

The synergy between managers and OR specialists is pivotal. OR models are most effective when there's continuous two-way colloration, ensuring models are not only technically sound but also pratical, trusted, and aligned with strategic goals.

2. Why is model validation considered a critical step in the OR process? What are the possible consequences of skipping this step?

⇒ Model validation is a critical step in the OR process because it ensures that the model accurately represents the real-world system it is intended to analyze or optimize. Without validation, there's no gaurantee that the model's results are reliable, which can lead to poor decision-making and wasted resources.

Why Model Validation is Critical

1. Ensures Realism and Accuracy
Validation checks whether the model behaves similarly to the actual system. This includes verifying that the data, assumptions and logic used in the model are correct.

2. Builds Trust with stakeholders
Managers and decision makers are more likely to accept and use a model when they are confident it reflects reality.

Date
Page

3. Improve Decision Quality
A validated model leads to more accurate predictions, better strategies, and lower risks.

4. Detects Errors Early
Validations can uncover modeling mistakes, incorrect data, or unrealistic assumptions before implementation, saving time & money.

5. Supports Continuous Improvement.
Validating models with real outcomes allows for refinement, making the model more robust and adaptive over time.

Consequence of Skipping Model Validation.

1. Incorrect or Misleading Results.
Decisions based on on unvalidated model may lead to suboptimal or even harmful outcomes.
Ex:-
A transportation company uses an unvalidated route optimization model and ends up increasing fuel costs due to unaccounted-for traffic patterns.

2. loss of credibility
If the model performs poorly in pratice, future modeling efforts may be met with skepticim or resistance.

Date
Page

3. Financial Loss

Implementing flawed strategies derived from invalid models can lead to wasted resources, lost revenue, or missed opportunities.

4. Operational Disruption

In systems like healthcare or supply chains, a faulty model can disrupt services, delay deliveries or cause safety issue.

5. Inability to Generalize.

A non-validated model may only work under specific, untested conditions and fail when used in different scenarios or over time.

Model validation is not optional- it's fundamental step that ensures an OR model is accurate, credible, and useful. Skipping this step turns the model from a decision-support tool into a source of risk.

A furniture manufacturer make two types of sofas. sofa of type A and sofa of type B. For simplicity, divide the production process into three distint operations, say carpentary, finishing and upholstery.

The amount of labour required for each operation varies. Manufacture of a sofa of type A requires 6 hrs of carpentary, 1 hrs of finishing & 2hrs of upholstery. Manufacture of sofa B requires 3hrs of capacity, 1 hrs of finishing and 6 hrs of upholstery. Owing to limited availability of skilled labour as well as of tools and equipment, the factory has available each day 96 man hrs of carpentary, 18 man hrs for finishing and 72 man hrs for upholstery.

The profit per sofa of type A is Rs 80 and the profit for sofa B is Rs 70. How many sofa of type A & B should be produced each day in order to maximize the profit? Formulate the problems as LPP.

Sol'n

Decision variable :-

x = no. of type A sofa per day

y = no. of type B sofa /day.

Objective function:

Maximize $Z = 80x + 70y$

Constraints

$$6x + 3y \le 96 \quad \text{------ (i)}$$

$$x + y \le \cancel{7}\ 18 \quad \text{------ (ii)}$$

$$2x + 6y \le 72 \quad \text{------ (iii)}$$

$$x, y \ge 0$$

In eq^n (i)

x	0	16
y	32	0

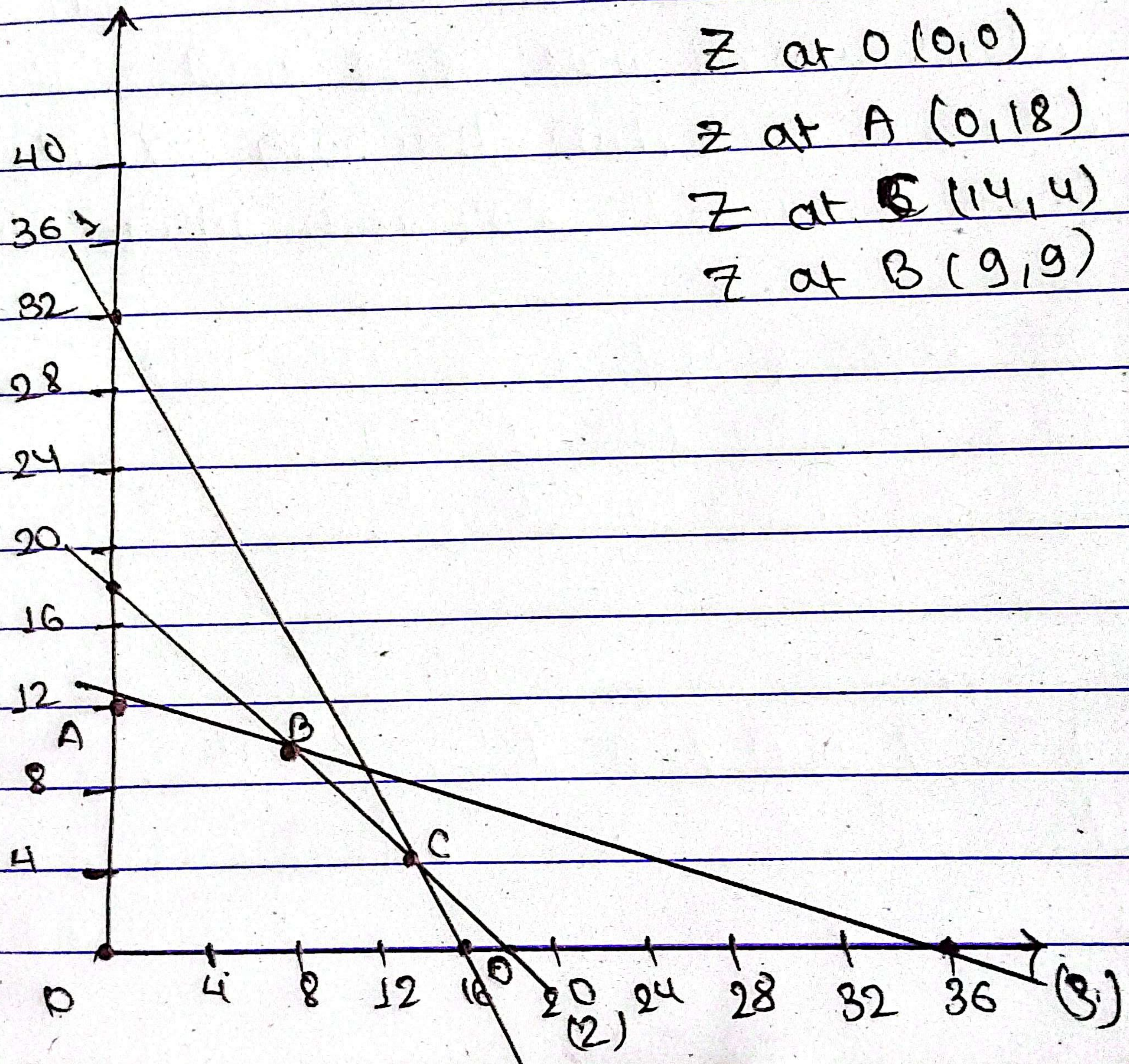
In eq^n ii

x	0	18
y	18	0

In eq^n iii

x	0	36
y	12	0

Z at C



Z at O (0,0) = 0

Z at A (0,18) = 1260

Z at C (14,4) = 1400 ✓

Z at B (9,9) = 1350

Russell Ackoff

Visual timeline showing the evolution of OR

- 1940s - Origins in WW II
  - Development of OR in millitary operations during world war II.
  - British military used OR to improve radar, anti-submarine warfare, and resource allocation
  - Patrick Blackett (UK Physicist) : led teams applying scientific method to warfare.

Outcome : Birth of formal OR

- 1950s - Post ware Expansion
  - OR moves into business and industry.
  - Linear programming and optimization techniques develop.
  - George Dantzing : Invented the simplex Method for linear programming (1947).

OR used in logistics, manufacturing, and inventory control.

- 1960s - 1970s - Academic and Institutional Growth.
  - Universities begin offering OR degrees.
  - Growth in computer technology aids modeling and simulations
  - Russel Ackoff : major figure in system thinking and OR in management.

Fields : Queueing theory, game theory, simulation models emerge.

- 1975 - Formation of INFORMS Predecessors
  - Operations Research Society of America (ORSA) and The Institute of Management Science (TIMS) dominate the field.
  - Foundations for the modern professional OR community.

- 1995 - Founding of INFORMS
  - Merger of ORSA and TIMS forms INFORMS (Institute of Operations Research and the Management Sciences)

Mission : Advance the development and use of OR and analytics.

- 2000s - Digital Optimization and Global Applications
  - OR techniques used in airline scheduling, supply chains, finance, and healthcare.
  - Expansion of OR software (e.g. CPLEX, Gurobi).

Ralph Gomory : for integer programming and cutting plane methods.

- 2010s - AI and OR Integration
  - Growth in machine learning, data science.
  - OR Integrates with AI for decision support, predictive analytics.

Tech : Reinforcement learning, neural network, simulation-based optimization.

Michael Trick : known for OR in sports scheduling and computational optimization.

- 2020s - Modern OR in Big Data & Decision Science.
  - OR applied to climate change, disaster relief, pandemic response.
  - Use of cloud computing and big data analytics.

Integration : Deep link between OR, AI and Operations Analytics.